

COMPOUND DISTRIBUTIONS ON THE WISHART MATRIX

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SUMMARY

In this paper a random matrix having a Wishart distribution where *a priori* the parameter matrix also has a Wishart distribution is considered. The resulting compound distribution is expressed in terms of a Bessel function of the second kind with matrix argument. Approximated and nonexplicit results for the case where the parameter matrix has a noncentral Wishart distribution are also given. Furthermore, the case where *a priori* a sub-matrix of the parameter matrix has a Wishart distribution is considered. Some properties of the determinant of a random matrix where the random matrix has the compound distributions mentioned above are discussed.

1. Compounding two Wishart distributions

The Bessel function of the second kind with matrix argument is defined as (Herz (1955), p. 506)

$$(1.1) B_{\gamma}(WZ)|W|^{\gamma} = \int_{\Lambda > 0} e^{-\text{tr} \Lambda W} e^{-\text{tr} \Lambda^{-1} Z} |\Lambda|^{-\gamma - \frac{1}{2}(p+1)} d\Lambda.$$

This integral is a Laplace transform in W and absolutely convergent for all γ provided $\text{Re}(W), \text{Re}(Z) > 0$. Changing variables from Λ to Λ^{-1} , it follows that

$$B_{-\gamma}(Z) = B_{\gamma}(Z)|Z|^{\gamma}.$$

Now let the conditional distribution of the random matrix X : $p \times p$ be a Wishart distribution with probability density function (p.d.f.) (Anderson (1958)).

$$(1.2) f_1(X|\Sigma_1) = \frac{|X|^{\frac{1}{2}(n-p-1)} e^{-\frac{1}{2}\text{tr} \Sigma_1^{-1} X}}{\Gamma_p(\frac{1}{2}n) 2^{\frac{1}{2}n} |\Sigma_1|^{\frac{1}{2}n}}, \quad \begin{matrix} X : p \times p > 0 \\ \Sigma_1 : p \times p > 0 \\ n > p - 1 \end{matrix}$$

where *a priori* the covariance matrix Σ_1 : $p \times p$ is assumed also to have a Wishart distribution with covariance matrix Σ_2 : $p \times p$ and p.d.f.

$$(1.3) f_2(\Sigma_1|\Sigma_2) = \frac{|\Sigma_1|^{\frac{1}{2}(m-p-1)} e^{-\frac{1}{2}\text{tr} \Sigma_2^{-1} \Sigma_1}}{\Gamma_p(\frac{1}{2}m) 2^{\frac{1}{2}m} |\Sigma_2|^{\frac{1}{2}m}}, \quad \begin{matrix} \Sigma_1 : p \times p > 0 \\ \Sigma_2 : p \times p > 0 \\ m > p - 1 \end{matrix}$$

From (1.2) and (1.3) the joint p.d.f. of the random matrix $X : p \times p$ and the random matrix parameter $\Sigma_1 : p \times p$ is given by

$$\begin{aligned} g(X, \Sigma_1 | \Sigma_2) &= f_1(X | \Sigma_1) f_2(\Sigma_1 | \Sigma_2) \\ &= \frac{2^{-\frac{1}{2}np} |X|^{\frac{1}{2}(n-p-1)} |\Sigma_1|^{\frac{1}{2}(m-n-p-1)}}{\Gamma_p(\frac{1}{2}n) \Gamma_p(\frac{1}{2}m) |2\Sigma_2|^{\frac{1}{2}m}} e^{-\frac{1}{2}\text{tr}\Sigma_1^{-1}X} e^{-\frac{1}{2}\text{tr}\Sigma_2^{-1}\Sigma_1}. \end{aligned}$$

According to (1.1) $h_1(X | \Sigma_2)$ can be written in terms of a Bessel function of the second kind as

$$\begin{aligned} (1.4) \quad h_1(X | \Sigma_2) &= \int_{\Sigma_1 > 0} g(X, \Sigma_1 | \Sigma_2) d\Sigma_1 \\ &= \frac{2^{-np} |X|^{\frac{1}{2}(n-p-1)}}{\Gamma_p(\frac{1}{2}n) \Gamma_p(\frac{1}{2}m) |2\Sigma_2|^{\frac{1}{2}n}} B_{\frac{1}{2}(n-m)}(\frac{1}{2}\Sigma_2^{-1}X); \end{aligned}$$

Special case:

For $\frac{1}{2}n = \frac{1}{2}m = p = 1$ in (1.4) and in the same way as Bhattacharya (1966) we obtain the following Bessel function model of life-time specified by the p.d.f.

$$h_1(x; \sigma_2^2) = \frac{1}{2\sigma_2^2} K_0\left(\frac{\sqrt{x}}{\sigma_2}\right) \text{ where } K_\gamma(y) = \frac{1}{2} B_\gamma(\frac{1}{2}y^2)(\frac{1}{2}y)^\gamma \text{ (Herz (1955)) and the}$$

failure rate is given by

$$Z(t; \sigma_2^2) = \frac{K_0\left(\frac{\sqrt{t}}{\sigma_2}\right)}{2\sigma_2 \sqrt{t} K_1\left(\frac{\sqrt{t}}{\sigma_2}\right)}.$$

Based on a sample $x_1, \dots, x_n$ from this distribution the moment estimate of σ_2^2 is equal to $\frac{1}{2}\mu'_1$ with $\mu'_1 = \frac{1}{n} \sum_{i=1}^n x_i$ the sample mean, and this is an unbiased estimator.

Remarks

(i) Using the approximation for the noncentral Wishart by a central Wishart in (1.3) with Σ_2 replaced by $\Sigma_2 + \frac{1}{n} \sum_{\alpha} \mu_{\alpha} \mu'_{\alpha}$ (Stein and Roux (1972)), $\mu_{\alpha} : p \times 1 = (\mu_{1\alpha}, \dots, \mu_{p\alpha})'$ is the mean vector from the appropriate $N(\mu_{\alpha}; \Sigma_2)$ distribution, and the $\mu_{j\alpha}$'s are small compared to the variances, then

$$h_2(X|\Sigma_2, \mu_\alpha) = \frac{2^{-np} |X|^{\frac{1}{2}(n-p-1)}}{\Gamma_p(\frac{1}{2}n) \Gamma_p(\frac{1}{2}m) |\Sigma_2 + \frac{1}{n} \sum_{\alpha} \mu_{\alpha} \mu'_{\alpha}|^{\frac{1}{2}n}} \\ \cdot B_{\frac{1}{2}(n-m)}(\frac{1}{2}(\Sigma_2 + \frac{1}{n} \sum_{\alpha} \mu_{\alpha} \mu'_{\alpha})^{-1} X).$$

(ii) Using the noncentral Wishart (Constantine (1963)), with matrix of noncentrality parameters $\Omega = \frac{1}{2} \sum_{\alpha} \mu_{\alpha} \mu'_{\alpha} \Sigma_2^{-1}$, in (1.3), then

$$(1.5) \quad h_3(X|\Sigma_2, \Omega) = \frac{2^{-\frac{1}{2}(m+n)p} e^{-\text{tr} \Omega} |X|^{\frac{1}{2}(n-p-1)}}{\Gamma_p(\frac{1}{2}n) \Gamma_p(\frac{1}{2}m) |\Sigma_2|^{\frac{1}{2}m}} \int_{\Sigma_1 > 0} |\Sigma_1|^{\frac{1}{2}(m-n-p-1)} \\ \cdot e^{-\frac{1}{2} \text{tr} \Sigma_1^{-1} X} e^{-\frac{1}{2} \text{tr} \Sigma_2^{-1} \Sigma_1} {}_0F_1(\frac{1}{2}m; \frac{1}{2} \Sigma_2^{-1} \Omega \Sigma_1) d\Sigma_1.$$

Although this density cannot be written in an explicit form, it will be used to derive some properties of $|X|$.

2. The decomposition of the matrices

Partition the random matrices $\Sigma_1 : p \times p$ and $X : p \times p$ as follows

$$\Sigma_1 = \begin{bmatrix} \Sigma_{11} : p_1 \times p_1 & \Sigma_{12} : p_1 \times p_2 \\ \Sigma_{21} : p_2 \times p_1 & \Sigma_{22} : p_2 \times p_2 \end{bmatrix} \\ X = \begin{bmatrix} X_{11} : p_1 \times p_1 & X_{12} : p_1 \times p_2 \\ X_{21} : p_2 \times p_1 & X_{22} : p_2 \times p_2 \end{bmatrix} \quad \text{with } p_1 + p_2 = p.$$

Let the condition distribution of the random matrix $X : p \times p$ be Wishart and let the marginal distribution of the random matrix $\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} : p_1 \times p_1$ conditional on Σ_{12} and Σ_{22} be Wishart with p.d.f.'s

$$(2.1) \quad f_1(X|\Sigma_1) = \frac{|X|^{\frac{1}{2}(n-p-1)} e^{-\frac{1}{2} \text{tr} \Sigma_1^{-1} X}}{\Gamma_p(\frac{1}{2}n) |2\Sigma_1|^{\frac{1}{2}n}}; \quad \begin{array}{l} X : p \times p > 0 \\ \Sigma_1 : p \times p > 0 \\ \Sigma_{22} : p_2 \times p_2 > 0 \\ n > p - 1 \end{array}$$

and

$$(2.2) f_3(\Sigma_{11.2}|\Sigma_3) = \frac{|\Sigma_{11.2}|^{1/2}(m_1-p_1-1) e^{-1/2 \text{tr} \Sigma_3^{-1} \Sigma_{11.2}}}{\Gamma_{p_1}(\frac{1}{2}m_1) |2\Sigma_3|^{1/2} m_1}; \quad \begin{array}{l} \Sigma_3 : p_1 \times p_1 > 0 \\ m_1 > p_1 - 1 \end{array}$$

The determinant and inverse of Σ_1 are expressed as (Morrison (1967), p. 66)

$$(2.3) |\Sigma_1| = |\Sigma_{22}| \cdot |\Sigma_{11.2}|$$

and

$$(2.4) \Sigma_1^{-1} = \begin{bmatrix} \Sigma_{11.2}^{-1} & -\Sigma_{11.2}^{-1} \Sigma_{12} \Sigma_{22}^{-1} \\ -\Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11.2}^{-1} & \Sigma_{22}^{-1} + \Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11.2}^{-1} \Sigma_{12} \Sigma_{22}^{-1} \end{bmatrix}$$

From (1.1), (2.1), (2.2), (2.3) and (2.4) $h_2(X|\Sigma_{12}, \Sigma_{22}, \Sigma_3)$ can be written in terms of a Bessel function of the second kind as

$$\begin{aligned} h_2(X|\Sigma_{12}, \Sigma_{22}, \Sigma_3) &= \int_{\Sigma_{11.2} > 0} f_1(X|\Sigma_1) f_3(\Sigma_{11.2}|\Sigma_3) d\Sigma_{11.2} \\ &= \frac{2^{-1/2} n p_1 |X|^{1/2} (n-p-1) e^{-1/2 \text{tr} \Sigma_{22}^{-1} X_{22}}}{\Gamma_p(\frac{1}{2}n) \Gamma_{p_1}(\frac{1}{2}m_1) |2\Sigma_{22}|^{1/2} |2\Sigma_3|^{1/2} n} B_{1/2(n-m_1)}(\frac{1}{2} \Sigma_3^{-1} W) \end{aligned}$$

with $W = X_{11} - \Sigma_{12} \Sigma_{22}^{-1} X_{21} - X_{12} X_{22}^{-1} \Sigma_{21} + \Sigma_{12} \Sigma_{22}^{-1} X_{22} \Sigma_{22}^{-1} \Sigma_{21}$.

3. Properties

The moments and the distribution of $|X|$, where the random matrix $X : p \times p$ has the distribution given in (i) (1.4) and (ii) (1.5) will now be discussed.

(a) Moments of $|X|$

The h -th moment of $|X|$, where the random matrix $X : p \times p$ has a p.d.f. $h(X)$ is defined as

$$E|X|^h = \int_{X > 0} |X|^h h(X) dX.$$

(i) From (1.1) by interchanging the order of integration, the h -th moment of $|X|$ is

$$(3.1) E_1 |X|^h = \frac{2^{2hp} \Gamma_p(\frac{1}{2}n+h) \Gamma_p(\frac{1}{2}m+h)}{\Gamma_p(\frac{1}{2}n) \Gamma_p(\frac{1}{2}m)} |\Sigma_2|^h.$$

(ii) From (1.5), by interchanging the order of integration, the h -th moment of $|X|$ follows as

$$(3.2) \quad E_2|X|^h = \frac{2^{2hp} \Gamma_p(\frac{1}{2}n+h) \Gamma_p(\frac{1}{2}m+h)}{\Gamma_p(\frac{1}{2}n) \Gamma_p(\frac{1}{2}m)} |\Sigma_2|^h \cdot {}_1F_1(-h; \frac{1}{2}m; -\Omega).$$

(b) *The distribution of $|X|$*

(i) From the definition of Meijer's G -function (Erdélyi (1953) and Mathai (1971)), the definition of the inverse Mellin transform (Erdélyi (1954)) and (3.1) the p.d.f. of $v = |X|$ follows as

$$(3.3) \quad f(v) = \frac{v^{-1} \pi^{\frac{1}{2}p(p-1)}}{\Gamma_p(\frac{1}{2}n) \Gamma_p(\frac{1}{2}m)} G_{0 \quad 2p}^{2p \quad 0} \left[2^{-2p} v |\Sigma_2|^{-1} \middle| b_1, \dots, b_{2p} \right]$$

$$\text{with } b_i = \begin{array}{ll} \frac{1}{2}n - \frac{1}{2}(i-1) & ; \quad i=1, 3, \dots, 2p-1 \\ \frac{1}{2}m - \frac{1}{2}(i-2) & ; \quad i=2, 4, \dots, 2p \end{array}$$

where $G_{0 \quad 2p}^{2p \quad 0}$ is Meijer's G -function.

Special case:

For $p=1$ and $x_{11} = x$ in (3.2) it follows that (Erdélyi, A. (1953))

$$f(x) = \frac{2^{-\frac{1}{2}(m+n-2)} \sigma_2^{-\frac{1}{2}(m+n)}}{\Gamma(\frac{1}{2}n) \Gamma(\frac{1}{2}m)} x^{\frac{1}{2}(m+n-4)} K_{\frac{1}{2}(n-m)} \left(\sqrt{\frac{x}{\sigma_2^2}} \right); \quad \begin{array}{l} 0 < x < \infty \\ n > 0 \\ m > 0 \end{array}$$

where $E(x) = nm\sigma_2^2$

and $\text{var}(x) = 2nm(n+m+2)\sigma_2^4$.

(ii) From (3.2) we see that

$$E_2|X|^h = E|Y_1|^h E|Y_2|^h$$

where Y_1 has a Wishart p.d.f. with covariance matrix I_p , Y_2 has a noncentral Wishart p.d.f. with covariance matrix Σ_2 , and Y_1 and Y_2 are independent.

Since the distribution of $|X|$ is of finite range, and is therefore uniquely determined by its moments, we have that the distribution of $|X|$ is the distribution of $|Y_1||Y_2|$.

Furthermore with the aid of (3.1) the h -th moment of $|X|$, given in (3.2), can be approximated as follows

$$\begin{aligned}(2.4) \quad E_2 |X|^h &= E_1 |X|^h |I + \frac{2}{m} \Omega|^{-h} + O\left(\frac{1}{m^3}\right) \\ &= E_1 |X|^h + \frac{2h}{m} \text{tr} \Omega E_1 |X|^{h-1} + O\left(\frac{1}{m^2}\right).\end{aligned}$$

Multiplying (3.4) by $\frac{\alpha^h}{h!}$ and taking the sum over all values of h we find that

$$M_2(\alpha) = M_1(\alpha) + \frac{2}{m} \text{tr} \Omega \alpha \frac{d}{d\alpha} M_1(\alpha) + O\left(\frac{1}{m^2}\right).$$

where $M_1(\alpha)$ and $M_2(\alpha)$ are the moment generating functions of $|X|$ respectively. This result was first proved by Steyn (1975) for the central and noncentral distribution of the determinant of sums of squares and products.

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